1 A spring is suspended from a clamp. Fig. 1.1 shows a pointer attached to the lower end of the spring.

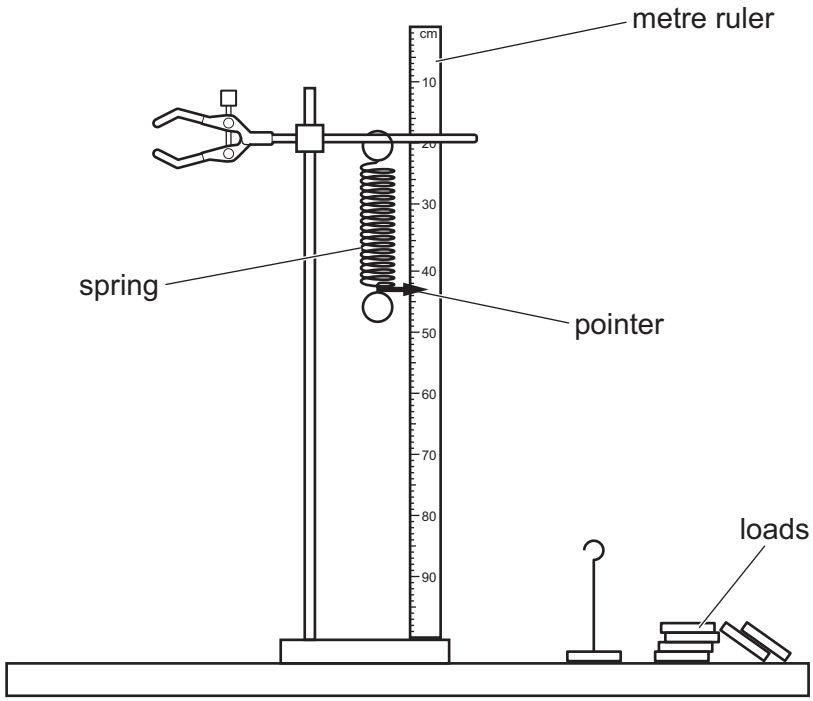

Fig. 1.1

A student suspends loads of different weights from the spring and records the readings on the metre ruler.

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Fig. 1.2 is the reading–weight graph that the student obtains.

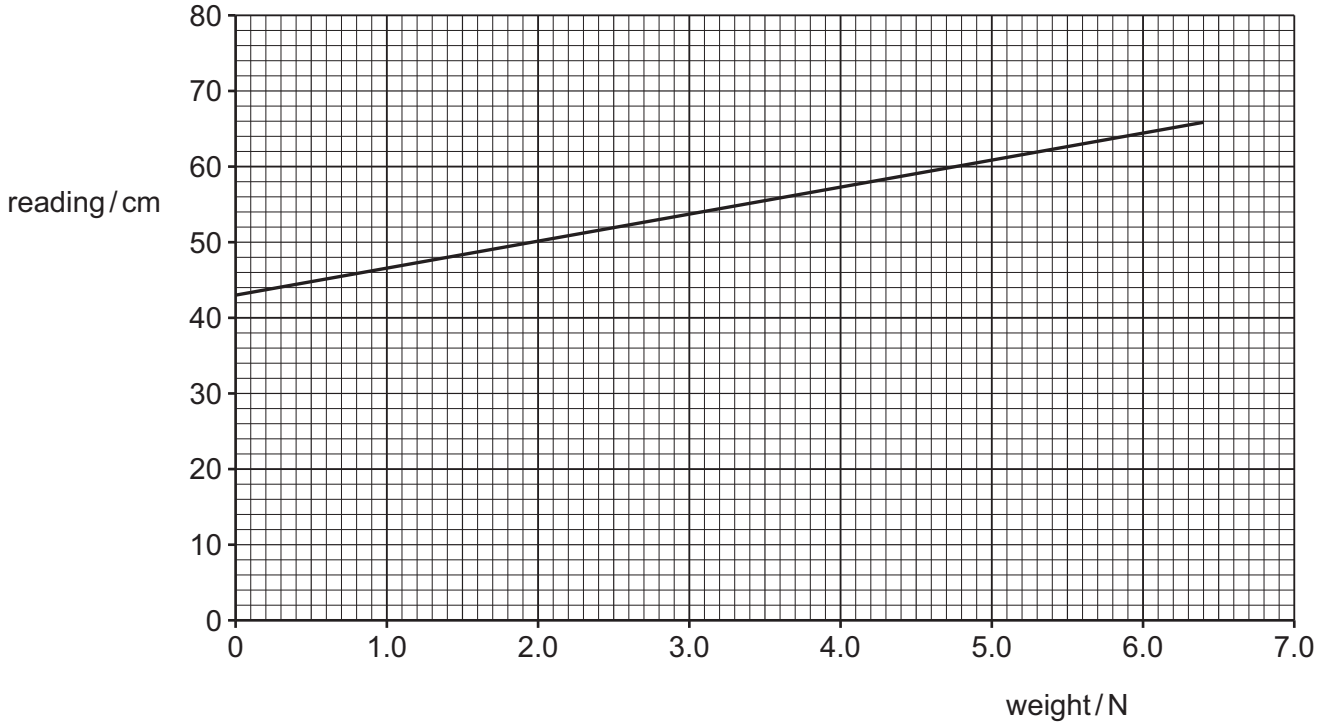

Fig. 1.2

(a) (i) Using Fig. 1.2, determine the reading on the metre ruler when

1. no weight is attached to the spring .....

2. a weight of 5.6 N is attached to the spring .....

[1]

(ii) Calculate the **extension** of the spring when the weight attached is 5.6 N.

extension = ... [1]

(b) Using the values found in **(a)**, calculate the spring constant of the spring.

spring constant = ... [2]

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(c) An object of mass 0.50 kg is attached to the spring.

(i) Calculate the weight of the object.

weight = ... [1]

(ii) The object is pulled downwards until the tension in the spring is 6.5 N.

The object is released.

Calculate the acceleration of the object immediately after it is released.

acceleration = ... [3]

[Total: 8]

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